# Linearized Barotropic Potential-Vorticity Equation in Mercator Coordinates

## 1 Mercator Coordinates

Let

$$x = \lambda, \qquad y = \log\tan\left(\frac{\pi}{4} + \frac{\phi}{2}\right), \qquad \frac{d\phi}{dy} = \cos\phi.$$

In these Mercator coordinates, the spherical metric is

$$ds^2 = a^2 \cos^2\phi\,(dx^2 + dy^2),$$

where $a$ is the radius of the sphere.

The computational domain is taken to be all longitudes and latitudes from 80°S to 80°N:

$$0 \le \lambda < 2\pi, \qquad -\frac{80\pi}{180} \le \phi \le \frac{80\pi}{180}.$$

Equivalently,

$$0 \le x < 2\pi, \qquad y_S \le y \le y_N,$$

where

$$y_S = \log\tan\left(\frac{\pi}{4} - \frac{80\pi}{360}\right), \qquad y_N = \log\tan\left(\frac{\pi}{4} + \frac{80\pi}{360}\right).$$

Since the Mercator coordinate is antisymmetric about the equator,

$$y_S = -y_N.$$

## 2 Perturbation Stream Function and Vorticity

Let $\psi(x, y, t)$ be the perturbation stream function. The perturbation velocity components are

$$u' = -\frac{1}{a\cos\phi}\psi_y, \qquad v' = \frac{1}{a\cos\phi}\psi_x.$$

The perturbation relative vorticity is

$$\zeta' = \nabla^2\psi = \frac{1}{a^2\cos^2\phi}\left(\psi_{xx} + \psi_{yy}\right).$$

Define the Mercator Laplacian

$$\Delta_M = \partial_x^2 + \partial_y^2.$$

Then

$$\zeta' = \frac{1}{a^2\cos^2\phi}\Delta_M\psi.$$

## 3 Background Flow and Background PV

The background flow is prescribed by the physical zonal and meridional winds

$$U = U(\lambda, \phi), \qquad V = V(\lambda, \phi).$$

In Mercator coordinates these are regarded as

$$U = U(x, \phi(y)), \qquad V = V(x, \phi(y)).$$

The Coriolis parameter is

$$f = 2\Omega \sin\phi.$$

The background absolute vorticity is

$$Q_0 = f + \zeta_0,$$

where the background relative vorticity is

$$\zeta_0 = \frac{1}{a\cos\phi}\left[V_x - \frac{1}{\cos\phi}\partial_y(U\cos\phi)\right].$$

Equivalently,

$$\zeta_0 = \frac{1}{a\cos\phi}\left(V_x - U_y + U\sin\phi\right).$$

The background material derivative is

$$\mathcal{D}_0 = \partial_t + \frac{U}{a\cos\phi}\partial_x + \frac{V}{a\cos\phi}\partial_y.$$

Define the Jacobian

$$J(A, B) = A_x B_y - A_y B_x.$$

## 4 Linearized Barotropic PV Equation

The inviscid, unforced linearized barotropic potential-vorticity equation for the perturbation stream function is

$$\mathcal{D}_0\left[\frac{1}{a^2\cos^2\phi}\Delta_M\psi\right] + \frac{1}{a^2\cos^2\phi}J(\psi, Q_0) = 0.$$

Now include linear friction proportional to minus the perturbation relative vorticity,

$$-r\nabla^2\psi = -r\zeta',$$

and include an external forcing $F(x, y, t)$. The forced-damped linearized equation is

$$\boxed{\mathcal{D}_0\left[\frac{1}{a^2\cos^2\phi}\Delta_M\psi\right] + \frac{1}{a^2\cos^2\phi}J(\psi, Q_0) = -r\left[\frac{1}{a^2\cos^2\phi}\Delta_M\psi\right] + F.}$$

Equivalently,

$$\boxed{\left(\partial_t + \frac{U}{a\cos\phi}\partial_x + \frac{V}{a\cos\phi}\partial_y\right)\left[\frac{1}{a^2\cos^2\phi}\left(\psi_{xx} + \psi_{yy}\right)\right] + \frac{1}{a^2\cos^2\phi}\left(\psi_x Q_{0y} - \psi_y Q_{0x}\right) = -r\left[\frac{1}{a^2\cos^2\phi}\left(\psi_{xx} + \psi_{yy}\right)\right]}$$

If the background flow is not itself a steady solution of the nonlinear barotropic potential-vorticity equation, then an additional inhomogeneous term remains:

$$\boxed{\mathcal{D}_0\left[\frac{1}{a^2\cos^2\phi}\Delta_M\psi\right] + \frac{1}{a^2\cos^2\phi}J(\psi, Q_0) = -r\left[\frac{1}{a^2\cos^2\phi}\Delta_M\psi\right] + F - \mathcal{D}_0 Q_0.}$$

## 5 Gaussian Forcing in the Niño 3.4 Region

The external forcing $F$ is chosen to be a Gaussian function of longitude and latitude, localized in the Niño 3.4 region. The Niño 3.4 box is

$$5^\circ\mathrm{S} \le \phi \le 5^\circ\mathrm{N}, \qquad 170^\circ\mathrm{W} \le \lambda \le 120^\circ\mathrm{W}.$$

Since the longitude domain is written as $0 \le \lambda < 2\pi$, the same longitude range is

$$190^\circ\mathrm{E} \le \lambda \le 240^\circ\mathrm{E}.$$

In radians,

$$-\frac{5\pi}{180} \le \phi \le \frac{5\pi}{180}, \qquad \frac{190\pi}{180} \le \lambda \le \frac{240\pi}{180}.$$

The center of the box is

$$\phi_c = 0, \qquad \lambda_c = \frac{215\pi}{180}.$$

Equivalently, $\lambda_c = 145^\circ\mathrm{W} = 215^\circ\mathrm{E}$.

Define the Gaussian widths

$$\sigma_\lambda = \frac{25\pi}{180}, \qquad \sigma_\phi = \frac{5\pi}{180}.$$

Because longitude is periodic, define the wrapped longitudinal distance

$$d_\lambda(\lambda, \lambda_c) = \operatorname{atan2}\left(\sin(\lambda - \lambda_c), \cos(\lambda - \lambda_c)\right).$$

This gives the shortest signed angular distance between $\lambda$ and $\lambda_c$.

The forcing is prescribed as

$$\boxed{F(\lambda, \phi, t) = F_0\, S(t) \exp\left[-\frac{1}{2}\left(\frac{d_\lambda(\lambda, \lambda_c)^2}{\sigma_\lambda^2} + \frac{(\phi - \phi_c)^2}{\sigma_\phi^2}\right)\right]}.$$

Here $F_0$ is the forcing amplitude and $S(t)$ is a nondimensional temporal envelope. For a steady forcing,

$$S(t) = 1.$$

For an oscillatory forcing, one may take

$$S(t) = \cos(\omega t).$$

For a pulsed forcing, one may take

$$S(t) = \exp\left[-\frac{(t - t_c)^2}{2\sigma_t^2}\right].$$

In Mercator coordinates, since $x = \lambda$ and $\phi = \phi(y)$, the forcing is

$$\boxed{F(x, y, t) = F_0\, S(t) \exp\left[-\frac{1}{2}\left(\frac{d_x(x, x_c)^2}{\sigma_\lambda^2} + \frac{(\phi(y) - \phi_c)^2}{\sigma_\phi^2}\right)\right]},$$

where

$$x_c = \lambda_c = \frac{215\pi}{180},$$

and

$$d_x(x, x_c) = \operatorname{atan2}\left(\sin(x - x_c), \cos(x - x_c)\right).$$

## 6 Prognostic-Diagnostic Form

Define the perturbation potential vorticity, equivalently the perturbation absolute-vorticity anomaly in the barotropic constant-depth case, by

$$q = \zeta' = \frac{1}{a^2 \cos^2 \phi} \Delta_M \psi.$$

Then the forced-damped linearized equation becomes the prognostic equation

$$\boxed{\mathcal{D}_0 q + \frac{1}{a^2 \cos^2 \phi} J(\psi, Q_0) = -rq + F.}$$

Equivalently,

$$\boxed{q_t + \frac{U}{a \cos \phi} q_x + \frac{V}{a \cos \phi} q_y + \frac{1}{a^2 \cos^2 \phi} \left(\psi_x Q_{0y} - \psi_y Q_{0x}\right) = -rq + F.}$$

With the Niño 3.4 Gaussian forcing, this becomes

$$\boxed{q_t + \frac{U}{a \cos \phi} q_x + \frac{V}{a \cos \phi} q_y + \frac{1}{a^2 \cos^2 \phi} \left(\psi_x Q_{0y} - \psi_y Q_{0x}\right) = -rq + F_0\, S(t) \exp\left[-\frac{1}{2}\left(\frac{d_x(x, x_c)^2}{\sigma_\lambda^2} + \frac{(\phi(y) - \phi_c)^2}{\sigma_\phi^2}\right)\right]}.$$

The diagnostic relation between $q$ and $\psi$ is the elliptic equation

$$\boxed{\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi = q.}$$

Equivalently,

$$\boxed{\Delta_M \psi = a^2 \cos^2 \phi\, q.}$$

## 7 Two-Step Time Advancement

Suppose $q^n$ and $\psi^n$ are known at time $t^n$. A simple explicit time discretization from $t^n$ to $t^{n+1} = t^n + \Delta t$ is as follows.

### 7.1 Step 1: Advance the PV

Use the prognostic equation to advance $q$:

$$\boxed{\frac{q^{n+1} - q^n}{\Delta t} + \frac{U}{a \cos \phi} q_x^n + \frac{V}{a \cos \phi} q_y^n + \frac{1}{a^2 \cos^2 \phi} J(\psi^n, Q_0) = -rq^n + F^n.}$$

Thus,

$$\boxed{q^{n+1} = q^n - \Delta t \left[\frac{U}{a \cos \phi} q_x^n + \frac{V}{a \cos \phi} q_y^n + \frac{1}{a^2 \cos^2 \phi} J(\psi^n, Q_0)\right] + \Delta t \left[-rq^n + F^n\right].}$$

For the Niño 3.4 Gaussian forcing,

$$\boxed{F^n(x, y) = F_0\, S(t^n) \exp\left[-\frac{1}{2}\left(\frac{d_x(x, x_c)^2}{\sigma_\lambda^2} + \frac{(\phi(y) - \phi_c)^2}{\sigma_\phi^2}\right)\right]}.$$

If the background is not a steady solution, replace the right-hand side by

$$-rq^n + F^n - \mathcal{D}_0 Q_0.$$

### 7.2 Step 2: Solve the Diagnostic Equation for the Stream Function

After $q^{n+1}$ has been computed, recover $\psi^{n+1}$ by solving

$$\boxed{\frac{1}{a^2 \cos^2 \phi} \Delta_M \psi^{n+1} = q^{n+1}.}$$

Equivalently,

$$\boxed{\Delta_M \psi^{n+1} = a^2 \cos^2 \phi \, q^{n+1}.}$$

Thus, at each time step, one first updates the prognostic PV variable $q$, and then solves the elliptic inversion problem for the stream function $\psi$.

## 8 Finite-Difference Formulation in Space and Time

Let the computational grid be

$$x_i = i\Delta x, \qquad i = 0, \ldots, N_x - 1, \qquad \Delta x = \frac{2\pi}{N_x},$$

and

$$y_j = y_S + j\Delta y, \qquad j = 0, \ldots, N_y - 1, \qquad \Delta y = \frac{y_N - y_S}{N_y - 1}.$$

The time grid is

$$t^n = t^0 + n\Delta t.$$

Write

$$q^n_{i,j} \approx q(x_i, y_j, t^n), \qquad \psi^n_{i,j} \approx \psi(x_i, y_j, t^n).$$

Define

$$\phi_j = \phi(y_j), \qquad m_j = a \cos \phi_j.$$

Thus

$$\frac{1}{a \cos \phi_j} = \frac{1}{m_j}, \qquad \frac{1}{a^2 \cos^2 \phi_j} = \frac{1}{m_j^2}.$$

The longitude direction is periodic. Therefore, for any grid field $A_{i,j}$,

$$A_{i+N_x,j} = A_{i,j}.$$

In particular,

$$A_{-1,j} = A_{N_x-1,j}, \qquad A_{N_x,j} = A_{0,j}.$$

The centered finite-difference approximations are

$$\delta_x A_{i,j} = \frac{A_{i+1,j} - A_{i-1,j}}{2\Delta x}, \qquad \delta_y A_{i,j} = \frac{A_{i,j+1} - A_{i,j-1}}{2\Delta y}.$$

The second derivatives are

$$\delta_{xx} A_{i,j} = \frac{A_{i+1,j} - 2A_{i,j} + A_{i-1,j}}{\Delta x^2}, \qquad \delta_{yy} A_{i,j} = \frac{A_{i,j+1} - 2A_{i,j} + A_{i,j-1}}{\Delta y^2}.$$

The discrete Mercator Laplacian is

$$\Delta^h_M A_{i,j} = \delta_{xx} A_{i,j} + \delta_{yy} A_{i,j}.$$

A centered approximation to the Jacobian is

$$J^h(A,B)_{i,j} = (\delta_x A)_{i,j}(\delta_y B)_{i,j} - (\delta_y A)_{i,j}(\delta_x B)_{i,j}.$$

The discrete diagnostic relation between $q$ and $\psi$ is

$$\boxed{q_{i,j}^n = \frac{1}{m_j^2}\Delta_M^h \psi_{i,j}^n.}$$

Equivalently,

$$\boxed{\Delta_M^h \psi_{i,j}^n = m_j^2 q_{i,j}^n.}$$

Written out explicitly,

$$\boxed{\frac{\psi_{i+1,j}^n - 2\psi_{i,j}^n + \psi_{i-1,j}^n}{\Delta x^2} + \frac{\psi_{i,j+1}^n - 2\psi_{i,j}^n + \psi_{i,j-1}^n}{\Delta y^2} = m_j^2 q_{i,j}^n.}$$

The background fields are evaluated on the grid as

$$U_{i,j} = U(x_i,\phi_j), \qquad V_{i,j} = V(x_i,\phi_j), \qquad Q_{0,i,j} = Q_0(x_i,y_j).$$

The Gaussian forcing is evaluated on the grid as

$$\boxed{F_{i,j}^n = F_0\, S(t^n) \exp\left[-\frac{1}{2}\left(\frac{d_x(x_i,x_c)^2}{\sigma_\lambda^2} + \frac{(\phi_j-\phi_c)^2}{\sigma_\phi^2}\right)\right].}$$

Here

$$\phi_c = 0, \qquad x_c = \frac{215\pi}{180}, \qquad \sigma_\lambda = \frac{25\pi}{180}, \qquad \sigma_\phi = \frac{5\pi}{180},$$

and

$$d_x(x_i,x_c) = \operatorname{atan2}\left(\sin(x_i-x_c),\cos(x_i-x_c)\right).$$

Using forward Euler in time and centered differences in space, the fully discrete PV update is

$$\boxed{\begin{aligned}\frac{q_{i,j}^{n+1}-q_{i,j}^n}{\Delta t} &+ \frac{U_{i,j}}{m_j}(\delta_x q^n)_{i,j} + \frac{V_{i,j}}{m_j}(\delta_y q^n)_{i,j} + \frac{1}{m_j^2}J^h(\psi^n,Q_0)_{i,j}\\ &= -r q_{i,j}^n + F_{i,j}^n.\end{aligned}}$$

Therefore,

$$\boxed{\begin{aligned}q_{i,j}^{n+1} = q_{i,j}^n &- \Delta t\left[\frac{U_{i,j}}{m_j}(\delta_x q^n)_{i,j} + \frac{V_{i,j}}{m_j}(\delta_y q^n)_{i,j} + \frac{1}{m_j^2}J^h(\psi^n,Q_0)_{i,j}\right]\\ &+ \Delta t\left[-r q_{i,j}^n + F_{i,j}^n\right].\end{aligned}}$$

Written out without operator notation, this is

$$\boxed{\begin{aligned} q_{i,j}^{n+1} = q_{i,j}^{n} - \Delta t &\left[ \frac{U_{i,j}}{m_j} \frac{q_{i+1,j}^{n} - q_{i-1,j}^{n}}{2\Delta x} + \frac{V_{i,j}}{m_j} \frac{q_{i,j+1}^{n} - q_{i,j-1}^{n}}{2\Delta y} \right] \\ - \Delta t &\left[ \frac{1}{m_j^2} \left( \frac{\psi_{i+1,j}^{n} - \psi_{i-1,j}^{n}}{2\Delta x} \frac{Q_{0,i,j+1} - Q_{0,i,j-1}}{2\Delta y} \right.\right. \\ &\qquad \left.\left. - \frac{\psi_{i,j+1}^{n} - \psi_{i,j-1}^{n}}{2\Delta y} \frac{Q_{0,i+1,j} - Q_{0,i-1,j}}{2\Delta x} \right) \right] \\ + \Delta t &\left[ -r q_{i,j}^{n} + F_0\, S(t^n) \exp\left[ -\frac{1}{2} \left( \frac{d_x(x_i, x_c)^2}{\sigma_\lambda^2} + \frac{(\phi_j - \phi_c)^2}{\sigma_\phi^2} \right) \right] \right]. \end{aligned}}$$

After this prognostic update, the stream function at the new time level is obtained from the discrete elliptic inversion

$$\boxed{\Delta_M^h \psi_{i,j}^{n+1} = m_j^2 q_{i,j}^{n+1}.}$$

That is,

$$\boxed{\frac{\psi_{i+1,j}^{n+1} - 2\psi_{i,j}^{n+1} + \psi_{i-1,j}^{n+1}}{\Delta x^2} + \frac{\psi_{i,j+1}^{n+1} - 2\psi_{i,j}^{n+1} + \psi_{i,j-1}^{n+1}}{\Delta y^2} = m_j^2 q_{i,j}^{n+1}.}$$

This is a linear elliptic problem for the unknowns $\psi_{i,j}^{n+1}$.

## 9 Boundary Conditions

The longitude direction is periodic:

$$q(0, y, t) = q(2\pi, y, t), \qquad \psi(0, y, t) = \psi(2\pi, y, t).$$

Equivalently, on the discrete grid,

$$q_{i+N_x,j}^{n} = q_{i,j}^{n}, \qquad \psi_{i+N_x,j}^{n} = \psi_{i,j}^{n}.$$

At the artificial northern and southern boundaries,

$$\phi = \pm 80^\circ,$$

one must impose a meridional boundary condition. Common choices are homogeneous Dirichlet conditions,

$$\psi = 0 \qquad \text{at} \qquad \phi = \pm 80^\circ,$$

or homogeneous Neumann conditions,

$$\frac{\partial \psi}{\partial y} = 0 \qquad \text{at} \qquad \phi = \pm 80^\circ.$$

For homogeneous Dirichlet conditions, the discrete boundary conditions are

$$\psi_{i,0}^{n+1} = 0, \qquad \psi_{i,N_y-1}^{n+1} = 0.$$

For homogeneous Neumann conditions, one may impose

$$\frac{\psi_{i,1}^{n+1} - \psi_{i,0}^{n+1}}{\Delta y} = 0, \qquad \frac{\psi_{i,N_y-1}^{n+1} - \psi_{i,N_y-2}^{n+1}}{\Delta y} = 0.$$

The corresponding treatment of $q$ at the meridional boundaries should be chosen consistently with the elliptic inversion for $\psi$.